

UST-OLOY, Russian Federation

WMO: 253250

Lat: 66.550N

Lon: 159.417E

Elev: 417

StdP: 14.48

Time Zone: 12.00 (E12)

Period: 90-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -55.6	(c) -53.0	(d) -59.9	(e) 0.2	(f) -55.3	(g) -57.3	(h) 0.2	(i) -53.0	(j) 16.6	(k) -12.4	(l) 13.1	(m) -11.5	(n) 2.7	(o) 230	(p) 0.810

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	79.4	61.7	75.0	59.0	70.9	57.0	63.1	75.1	60.6	72.5	58.2	68.7	4.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.3	73.7	68.4	55.5	66.5	64.2	52.9	60.5	62.0	28.9	75.0	27.1	72.7	25.4	68.9	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB
(a) 15.8	(b) 13.2	(c) 11.1	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	
(a) 15.8	(b) 13.2	(c) 11.1	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-33.7	-31.3	-13.7	8.8	36.3	53.1	56.9	50.0	37.4	10.8	-14.4	-31.8	
	DBStd	35.59	13.43	11.70	14.27	12.97	9.44	7.06	7.77	7.07	7.90	13.36	14.58	13.94	
	HDD50	14750	2596	2278	1976	1237	432	48	26	90	382	1215	1933	2537	
	HDD65	19765	3061	2698	2441	1687	889	359	272	465	828	1680	2383	3002	
	CDD50	482	0	0	0	0	8	142	238	90	4	0	0	0	
	CDD65	22	0	0	0	0	0	3	19	0	0	0	0	0	
	CDH74	474	0	0	0	0	0	91	346	37	0	0	0	0	
	CDH80	100	0	0	0	0	0	13	82	5	0	0	0	0	
Wind	WSAvg	3.5	2.1	2.1	3.2	3.6	5.0	4.8	4.3	4.4	4.4	3.3	2.3	2.5	
Precipitation	PrecAvg	10.2	0.6	0.5	0.4	0.3	0.5	1.2	1.9	1.7	1.0	0.8	0.8	0.6	
	PrecMax	14.4	2.2	1.5	1.1	1.3	2.0	4.2	4.3	4.1	3.4	2.0	2.0	1.7	
	PrecMin	6.6	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2	0.1	0.2	0.2	0.0	
	PrecStd	2.0	0.4	0.3	0.3	0.3	0.4	0.8	1.1	0.9	0.7	0.4	0.4	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	14.1	31.6	43.0	68.8	81.4	86.1	78.3	63.4	39.5	30.4	19.0	
		MCWB	14.7	12.9	26.1	34.9	50.8	59.7	64.5	61.2	49.9	36.3	26.5	18.6	
	2%	DB	-1.0	-1.0	23.8	38.8	62.1	76.3	82.1	72.8	58.7	34.6	17.9	5.2	
		MCWB	-1.9	-2.1	20.1	32.0	47.1	57.2	63.0	59.3	48.3	31.9	16.6	4.4	
	5%	DB	-7.5	-6.8	17.0	34.6	57.7	72.7	77.9	68.1	54.3	31.6	10.8	-2.4	
		MCWB	-8.1	-7.4	14.6	29.3	44.7	55.6	61.9	56.8	46.0	29.6	10.0	-3.0	
	10%	DB	-13.9	-11.4	9.3	30.4	52.7	68.9	73.2	63.8	49.6	28.5	5.1	-8.6	
		MCWB	-14.3	-11.9	7.6	26.1	42.2	53.7	59.1	54.8	44.5	26.2	4.2	-9.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	12.6	26.8	35.5	51.3	61.7	68.5	65.3	52.7	36.6	26.7	19.0
			MCDB	16.8	14.2	30.5	42.2	66.6	75.4	82.0	73.7	60.3	39.1	30.1	19.7
2%		WB	-1.4	-2.0	20.2	32.4	47.9	58.5	64.8	60.3	49.5	32.5	16.8	4.3	
		MCDB	-0.9	-1.1	23.5	38.1	60.9	72.2	78.2	70.5	55.9	33.9	17.9	5.1	
5%		WB	-8.0	-7.4	14.5	29.5	44.9	56.6	62.4	57.3	47.1	29.5	10.0	-3.1	
		MCDB	-7.5	-6.9	16.8	34.4	56.7	70.3	74.8	67.0	52.6	31.8	10.8	-2.5	
10%		WB	-14.2	-11.9	7.7	26.1	42.4	54.5	60.3	55.2	44.8	26.0	4.3	-9.1	
		MCDB	-13.8	-11.4	9.3	30.5	52.4	67.3	71.8	62.5	49.3	28.4	4.9	-8.5	
Mean Daily Temperature Range	5% DB	MDBR	12.0	15.7	23.1	24.9	18.3	20.4	20.3	18.3	13.7	13.7	13.3	10.1	
		MCDBR	17.3	18.0	28.3	21.3	25.0	29.5	30.5	30.5	24.5	14.6	19.9	15.0	
		MCWBR	16.9	17.3	25.0	15.2	13.7	12.7	14.3	16.7	14.5	12.4	18.7	14.5	
	5% WB	MCDBR	17.3	17.9	27.8	19.7	23.4	26.6	27.9	26.3	20.8	14.1	19.8	15.2	
		MCWBR	16.9	17.4	24.6	14.4	13.3	12.7	14.6	15.3	13.1	12.2	18.7	14.8	
Clear-Sky Solar Irradiance	taub	0.167	0.234	0.299	0.354	0.392	0.375	0.449	0.422	0.326	0.270	0.186	0.174		
	taud	1.917	2.089	2.048	1.945	2.097	2.357	2.128	2.160	2.400	2.159	1.916	1.907		
	Ebn at Noon	81	197	233	247	252	264	234	226	232	175	61	6		
	Edh at Noon	5	16	30	44	42	33	41	36	22	15	4	1		
All-Sky Solar Radiation	RadAvg	24	235	810	1579	1816	1771	1555	1171	636	251	52	2		
	RadStd	3	21	25	51	145	133	147	107	85	35	5	0		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page